

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Statistical Physics **L-T-P:** 3-1-0 **Credits:** 4

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Fundamental postulates; Phase space, Liouville's equation, ergodicity; Ensembles: Microcanonical, canonical and grand canonical ensembles; Partition functions; Connection with thermodynamics; Examples of non-interacting systems: Classical ideal gas, harmonic oscillators, paramagnetism, two-state systems; Equipartition; Negative temperature. Fluctuations and equivalence of ensembles. (16) Quantum statistical mechanics: Density matrix; Ensembles; Examples: An electron in a magnetic field, A free particle in a box; A linear harmonic oscillator; Ideal quantum gases: Bose-Einstein and Fermi-Dirac distribution; Bose-Einstein condensation; Photons and Phonons; Superfluidity; Fermi gas; Pauli paramagnetism and Landau diamagnetism; White dwarf stars. Gases with internal degrees of freedom. (16) Interacting systems: van der Waals equation of state; Cluster expansions; Phase transition in model systems. (6) Fluctuation; Dissipation and linear response; Langevin equation; Fokker-Planck equation. (4)

Reference Books:

1. Statistical Mechanics: R. K. Pathria and P. D. Beale
2. Fundamentals of Statistical and Thermal Physics: F. Reif
3. Statistical Mechanics: Kerson Huang
4. Statistical Physics: L.D. Landau and E.M. Lifshitz

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Fundamental postulates; Phase space, Liouville's equation, ergodicity; Ensembles: Microcanonical, canonical and grand canonical ensembles; Partition functions; Connection with thermodynamics; Examples of non-interacting systems: Classical ideal gas, harmonic oscillators, paramagnetism, two-state systems; Equipartition; Negative temperature. Fluctuations and equivalence of ensembles.	16
2.	Quantum statistical mechanics: Density matrix; Ensembles; Examples: An electron in a magnetic field, A free particle in a box; A linear harmonic oscillator; Ideal quantum gases: Bose-Einstein and Fermi-Dirac distribution; Bose-Einstein condensation; Photons and Phonons; Superfluidity; Fermi gas; Pauli paramagnetism	16

	and Landau diamagnetism; White dwarf stars. Gases with internal degrees of freedom.	
3.	Interacting systems: van der Waals equation of state; Cluster expansions; Phase transition in model systems.	6
4.	Fluctuation; Dissipation and linear response; Langevin equation; Fokker-Planck equation.	4
Total number of hours		42